

Global Support Programme

Land Degradation Neutrality Target Setting Programme

Methodological note to set national voluntary Land Degradation Neutrality (LDN) targets using the UNCCD indicator framework

June 2017

1. Introduction

This methodological note has been prepared based on the document developed during the UNCCD LDN pilot project and in the context of two Programmes implemented by the UNCCD Secretariat and the Global Mechanism in cooperation with multiple partners, i.e. the Global Support Programme and the Land Degradation Neutrality Target Setting Programme (LDN TSP) (herein referred to as the Programmes). The note aims at providing guidance to countries committed to set national voluntary LDN targets.

The UNCCD Conference of the Parties (COP) adopted the following indicators (and associated metrics) to report on progress in the implementation of the Convention and to monitor, evaluate and communicate progress towards achieving LDN targets:

- trends in land cover (vegetative land cover);
- trends in land productivity or functioning of the land (land productivity dynamics);
- trends in carbon stocks above and below ground (soil organic carbon (SOC) stock).

The COP requested the UNCCD secretariat, in cooperation with relevant specialized institutions, to:

- a) Compile and make available to affected country Parties national estimates of the metrics/proxies associated with these indicators from available global datasets as default data for validation in accordance with the procedure established in decision 22/COP.11;
- b) Prepare methodological guidelines and provide technical assistance to affected country Parties on the compilation and use of such default data;
- c) Undertake measures aimed at strengthening the capacities of affected Parties to validate, replace or reject the default data.

These activities are funded, among others, under the Global Support Programme, which aims at increasing the quantity and improving the quality of information for the review of implementation of the UNCCD through strengthening Parties' capacities in monitoring and reporting, knowledge management and outreach. The Global Support Programme is implemented in close cooperation with the UNCCD LDN TSP, which is supporting countries to establish national voluntary LDN targets and identifying associated measures to achieve these targets. As of 21 November 2016, 102 countries committed to set national LDN targets and expressed an interest in participating in the LDN TSP.

The UNCCD reporting and LDN target setting processes are contributing to achieving the Sustainable Development Goal (SDG) 15, Life on Land, and in particular its target 15.3¹ as adopted by the United Nations General Assembly (UNGA, 2015).

The UNCCD COP 12 endorsed SDG target 15.3 and the concept of LDN as a strong vehicle for driving the implementation of the Convention, agreed on a definition of LDN² and invited country Parties to set LDN national voluntary targets. The UNCCD Secretariat and the Global Mechanism are therefore implementing the above-mentioned support programmes to assist country Parties wishing to do so in setting LDN targets, including the definition of national baselines, measures and targets to achieve LDN by 2030.

This note complements the draft LDN target setting technical guide, which was disseminated among countries participating in the Programme at the occasion of the regional inception workshops,³ and the related brochure “Achieving Land Degradation at the country level: Building blocks for LDN Target Setting”,⁴ which explains in practical terms how to put the evolving LDN target setting process into practice. In particular, this note provides additional information on one of the 4 building blocks that form the basis for the LDN target setting process, i.e. building block 2 on “Assessing LDN”. Especially, the note provides information on the following:

- The minimum set of indicators recommended for tracking progress towards LDN targets (section 2);
- The default data derived from global data sources and made available by the Programmes for use and validation by countries in the absence of, or to complement and enhance, national data sources (section 3);
- The use of the LDN indicators to set the LDN baseline, assess land degradation trends, and monitor progress towards LDN targets (section 4).

2. The LDN indicators

The aspirational goal of LDN is to **maintain or enhance the natural capital of the land** and associated land-based ecosystem services. LDN therefore implies a commitment to **avoid further net loss** of the land-based natural capital relative to a reference state, or baseline.

The indicators recommended for tracking progress towards LDN are:

- **land cover**
- **land productivity**
- **carbon stocks (metric: soil organic carbon (SOC) stock)**

¹ “By 2030, combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land-degradation-neutral world” (UNCCD, 2015).

² “Land Degradation Neutrality is a state whereby the amount and quality of land resources, necessary to support ecosystem functions and services and enhance food security, remains stable or increases within specified temporal and spatial scales and ecosystems”.

³ UNCCD, 2016a. “Land Degradation Neutrality Target Setting. A technical guide. Draft for consultation during the Land Degradation Neutrality Target Setting Programme inception phase”.

⁴ UNCCD, 2016b. Achieving Land Degradation Neutrality at the country level, Building blocks for LDN target setting. Available at: http://www2.unccd.int/sites/default/files/documents/18102016_LDN%20country%20level_ENG.pdf

Together the three indicators provide good coverage of the condition of land-based natural capital and ecosystem services that are provided from that land base. In addition, they address change in the system in different yet highly relevant ways. Land cover provides a first indication of changing vegetation cover, to some extent as proxy of the underlying use, and of land conversion and resulting habitat fragmentation. Land productivity offers an indication of ecosystem functioning and health and sharpens the focus on ecosystem services. Carbon stocks, especially soil organic carbon stocks, denotes overall soil quality.⁵

However, they represent a minimum set that **could be enhanced and complemented by national (or subnational) level indicators** for a more accurate picture of land degradation.

3. The default data sets made available by the Programme

The indicators should be computed primarily, and to the largest extent possible, using comparable and standardized **national data sources**. Only in their absence or as a complement to national data **global data sources** should be used. While in the long term, all countries should ideally be able to perform relevant data collection, analysis and report on land degradation, independently, global data sources can bridge data gaps and ultimately decrease the reporting burden on countries.

A tiered approach is therefore recommended for the computation of the indicators:

- Tier 1 (default method): Global/regional earth observation, geospatial information and modelling;
- Tier 2: National statistics based on data acquired for administrative or natural reference units (e.g. watersheds) and national earth observation;
- Tier 3 (most detailed method): Field surveys, assessments and ground measurements.

This approach enables national authorities to use methods consistent with their capacities, resources and data availability and facilitates comparability at global level.

As part of the Programmes, countries participating in the LDN TSP are provided with default tier 1 data derived from global data sources for their use and validation in the absence of national data.⁶ While a number of suitable global data sources are available for the computation of the indicators, the selected sources of default tier 1 data were identified based

⁵ SOC is one of the most important constituents of soil due to its capacity to promote plant growth, recycle nutrients to maintain soil fertility, and clean and store freshwater whilst reducing downstream flooding and promoting dry season flows. SOC is therefore intrinsically connected to soil quality. Maintaining carbon stocks in soils by providing adequate fresh organic matter for decomposition (and/or preventing excessive loss) can also generate additional benefits pertaining to climate change mitigation and biodiversity conservation. A common point of all forms of land degradation is soil organic carbon content depletion, where reduced organic matter inputs and inappropriate use destroy soil structure and reduce biodiversity leading to the progressive erosion of the non-renewable mineral fraction of soil. Once degraded, this mineral fraction is difficult to impossible to recover in the near future (most mineral fractions of soils are far in excess of 10 000 years in development, weathering from solid rock into more biologically useful particle sizes). Soil organic carbon stocks are influenced by land-use and management activities that affect litter input rates and soil organic matter loss rates: inputs are primarily controlled by decisions impacting NPP and/or the retention of dead organic matter (e.g. how much harvested biomass is removed as products and how much is left as residues), while outputs are mostly influenced by management decisions that affect microbial and physical decomposition of soil organic matter (e.g. tillage intensity) (IPCC, 2006). Depending on interactions with previous land use, climate and soil properties, changes in management practices may induce increases or decreases in soil carbon stocks.

⁶ All country Parties will be provided with this default global data as part of the Global Support Programme in support of the UNCCD reporting process.

on considerations of immediate availability and readiness for use, as along with temporal and spatial coverage and resolution.

This section describes the default data which are made available by the Programmes in partnership with the European Space Agency (ESA), the Joint Research Centre of the European Commission (JRC) and the International Soil Reference and Information Centre (ISRIC). More information on the accuracy and limitations of the default data are contained in annex I. At the end of this section, some guidelines for countries who decide to use national datasets or alternative data sources are provided.

Default data are pre-processed by the Programmes for the analysis of the indicators at the national level. Indicators will be also examined at watershed level, following the border of the major basin or sub basin boundary according to the morphological characteristics of the area and at the request of the countries. Further information aimed at guiding countries in the LDN analysis at the watershed level will be shared as an additional annex to this methodological note, as soon as the data processing will be finalized.

3.1. Default data on land cover

The ESA's Climate Change Initiative Land Cover dataset (CCI-LC)⁷ is used as default source of land cover data. The available version of the ESA-CCI-LC dataset (v. 1.6.1):

- encompasses three epochs centered around 2000, 2005 and 2010;⁸ and
- uses a hierarchical classification based on the Food and Agriculture Organization's Land Cover Classification Systems (LCCS), with 22 classes at 'level 1' for the entire world and 14 additional classes at "level 2" legend based on more accurate and regional information where available.

In order to obtain an indication of the **land use/cover baseline** for LDN target setting, the ESA CCI-LC 2000 epoch is used and the 22 original classes are aggregated into 6 main land cover categories as detailed in Table 1. These categories are selected because they are implementable, complete (in that all land areas in a country may be classified by these categories without duplication) and aligned with the 6 land use categories recommended by IPCC for the purpose of estimating the level of total carbon content for the global estimates of land-based GHG sinks and sources (IPCC, 2006). For the Programme, the land cover baseline data are made available to each country both in their original format (22 'level 1' classes) and as reclassified in 6 main land cover classes.

To obtain estimates of **land cover change**, the ESA CCI-LC 2000 and 2010 epochs are used, focusing on changes between the 6 main land cover categories only. While the ESA CCI-LC dataset was not made to provide land cover changes *per se*, the comparison of the two epochs generally allows for the identification of main land conversion trends. However the dataset shows some shortfalls on the urban and wetland layers, which will be addressed in the next release of the ESA-CCI-GLC project.⁹

⁷ More information on this data set can be found at: <http://www.esa-landcover-cci.org/>.

⁸ The next release of the ESA-CCI Land Cover project will consist of global land cover annual maps from 1992 to 2015. These annual maps will also provide the land cover changes at 300m since 2002 and at 1 km before 2002. As soon as validated, the complete dataset will be made available for the LDN TSP.

⁹ In order to improve the urban layer, the next release of the ESA-CCI Land Cover dataset will integrate in its processing chain the JRC's Global Human Settlement Layer (<http://ghsl.jrc.ec.europa.eu/>). For the wetland

The general file specifications for the country specific land cover data are as follows:

- Coordinate system: GCS_WGS_1984;
- Cell size (X, Y): 0,0027777777, 0,0027777777 [decimal degrees], corresponds to approximately 300 meters;
- Source: Data derived from ESA CCI-LC (2000, 2010) by reclassification from 23 full classes to 6 main aggregated classes;
- Digital class values: 1-6 (Table 1).

The land cover changes were coded following a 2 digit system: the first one is the class code for 2000 and the second is the class code for 2010. For example:

- 11, 22, 33, 44, 55 and 66 mean no change between land cover 2000 and 2010;
- 13 means a change from forest (code 1) to cropland (code 3) between 2000 and 2010;
- 45 means a change from wetland (code 4) to cropland (code 3) between 2000 and 2010.

layer, some improvements are expected as a result of the ESA's GlobWetland Africa, but the availability of an improved wetland layer in the ESA-CCI Land Cover dataset will require more time.

Table 1: Land use /cover categories

Value	Categories	Short description	ESA CCI-LC classes (codes)
1	Forests	Geographical areas dominated by natural tree plants with a cover of 15% or more. This class also includes: - mosaic tree and shrub (>50%) / herbaceous cover - seasonally or permanently tree flooded with fresh water	Tree broadleaved evergreen, Tree broadleaved deciduous, Tree needle leaved evergreen, Tree needle leaved deciduous, Tree mixed leaf type, Mosaic tree, shrub / herbaceous cover, Tree flooded, fresh water (50, 60, 61, 62, 70, 71, 72, 80, 81, 82, 90, 100, 160)
2	Shrubs, grasslands and sparsely vegetated areas	Geographical areas dominated by: - natural shrubs; or - natural herbaceous plants; or - sparse natural vegetation with a cover of 15% or less; This class also include: - mosaic natural vegetation (>50%) / crops - mosaic herbaceous cover (>50%) / tree and shrub	Mosaic natural vegetation / cropland, Mosaic herbaceous cover / tree, shrub, Scrublands, Grassland, Lichens and mosses, Sparse vegetation (40,110, 120, 121, 122, 130, 140, 150, 152, 153)
3	Cropland	Geographical areas dominated by: - herbaceous crops; or - woody crops; or - mixed herbaceous and woody crops; This class also include: - mosaic crops (50%) / natural vegetation	Cropland rainfed, Herbaceous cover Tree or shrub cover Cropland, irrigated or post-flooding, Mosaic cropland / natural vegetation (10, 11, 12, 20, 30)
4	Wetlands and water bodies	Geographical areas dominated by: - shrub or herbaceous vegetation, aquatic or regularly flooded; or - mangroves or - water bodies (natural/artificial, standing/flowing, inland/sea)	Tree cover, flooded, saline water, Shrub or herbaceous cover, flooded, fresh/saline/brakish water Water bodies (170,180,210)
5	Artificial areas	Geographical areas dominated by artificial surfaces , including urban and associated areas (e.g. urban parks), transport infrastructures, industrial areas, burnt areas, waste deposits, extraction sites.	Urban areas (190)
6	Bare land and other areas	Geographical areas dominated by : - bare areas or - snow and glaciers	Bare areas, Permanent snow and ice (200, 201, 202, 220)

3.2. Default data on land productivity

The JRC's Land Productivity Dynamics¹⁰ (LPD) dataset is used as default source for land productivity data (Cherlet et al. 2014, Annex 2 in Genesis et al. 2014). The LPD dataset is derived from a 15-year time series (1999 to 2013) of global NDVI observations composited in 10-day intervals at a spatial resolution of 1 km.¹¹

The LPD dataset provides 5 qualitative classes of land productivity trends over the above-mentioned time period as shown in table 2. These qualitative classes do not directly correspond to a quantitative measure (e.g. t/ha of NPP or GPP) of lost or gained biomass productivity, nevertheless there is an indirect relationship. The 5 classes are rather a qualitative combined measure of the intensity and persistence of negative or positive trends and changes of the photo-synthetically active vegetation cover over the observed period. While not an absolute measure of land productivity it depicts trajectories of long-term seasonal dynamics and departures from it that are typically related to overall land productivity change.

Given the nature of the LPD dataset, the **land productivity baseline** can in this case not be directly expressed as 'absolute numerical value' of GPP or NPP over a specific time period, but rather as a **baseline trend**. While using absolute numerical values rather than trends for monitoring LDN is preferable,¹² the LPD dataset describes the directness and persistence of negative or positive alterations of land productivity throughout the baseline period and is therefore useful for assessing land degradation, practical LDN target setting as well as for further monitoring.¹³

For LDN target setting, the LPD dataset is disaggregated by the 6 main land cover categories listed in Table 1, i.e. the statistical distribution of the 5 LPD classes within each of the 6 land cover categories is extracted.

¹⁰ The term "dynamics of land productivity" refers to the fact that the primary productivity of a stable land system is usually highly variable between different years/vegetation growth cycles as a function of natural (semi-natural systems) or partially human induced (e.g. SLM) adaptation and resilience to diverse environmental conditions and human intervention. Hence a land system's primary productivity assembles rather a dynamic equilibrium than a linear evenly evolving continuum.

¹¹ More information on input data and LPD processing chain concept can be found at: http://land.copernicus.eu/global/sites/default/files/products/GIOGL1_PUM_NDVIV1_I1.10.pdf and http://wad.jrc.ec.europa.eu/data/EPreports/LPDinEU_final_no-numbers.pdf respectively.

¹² For explanation of why absolute values rather than trends are used to monitor neutrality, see UNCCD/Science-Policy Interface (2016).

¹³ Given that the global time series of daily observations of vegetation indices such as the NDVI (or others) are continuously and operationally up-dated, for each monitoring step the extended NDVI time series will be used to produce again the LPD classes but with the then longer time series as input. Thus LPD class changes between the baseline period and the follow-up monitoring steps will indicate changes in land productivity trajectories. In parallel it is proposed to address the demand to address land productivity monitoring rather with numerical values of land productivity change than with 'qualitative classes' of land productivity dynamics, by providing information on percent change of land productivity between the baseline and each subsequent monitoring reference year (GPP proxy expressed as average of time integrated NDVI over a 3 to 5 years window around the baseline year and the monitoring reference years). In terms of maturity and "operational readiness", the estimation of GPP at national and sub-national levels, with spatial resolution between 1000 to 250m, the use of remote sensing inputs in the form of vegetation indexes, that reflect green vegetation cover dynamics and spatial heterogeneity at these scales, are currently most realistic and possible to use routinely (Yengoh et al. 2015). Extension of the LPD approach to 30 m resolution for specific areas using available Landsat archives and new data sources (e.g. Copernicus Sentinel) is becoming increasingly realistic.

Table 2: classes of land productivity dynamics

Value	Description
1	Declining productivity
2	Early signs of decline
3	Stable, but stressed
4	Stable, not stressed
5	Increasing productivity

The general file specifications of the global LPD data set are as follows:

- Coordinate system: GCS_WGS_1984
- Cell size (X, Y): 0,0089285714, 0,0089285714 [decimal degrees], corresponds to approximately 1 sqkm.
- Digital class values: 1-5

The general file specifications of country specific LPD data are as follows:

- Coordinate system : GCS_WGS_1984
- Cell size (X, Y): 0,0027777777, 0,0027777777 [decimal degrees], corresponds to approximately 300m. (Global LPD dataset (1 sqkm) has been converted in order for integration with the land cover dataset, at 300 m resolution.)
- Digital class values: 1-5

3.3. Default data on soil organic carbon (SOC) stocks

Populating the Soil Organic Carbon (SOC) indicator is challenging but data sources and methodologies for this vital indicator are now becoming available. In order to derive trends in SOC suitable for LDN target setting, two types of information are required:

1. the baseline SOC stocks (e.g. ton/ha) for the country in the year of interest (here 2000), and
2. some way of relating changing land use/cover conditions to changes in SOC stocks.

Baseline SOC stocks and their change can either be modeled or directly measured, or some combination thereof. In order to obtain an indication of default baseline SOC stocks, ISRIC's¹⁴ SoilGrids250m (Hengl et al., 2016) products of SOC percentage, bulk density, gravel fraction and depth to bedrock were used to calculate a predicted SOC stock for 0 – 30 cm (i.e. topsoil). Whilst SoilGrids250m was not made to represent the state of SOC in soils in the year 2000 (being constructed from soil data collected over several decades), it is globally consistent and readily accessible, contains the modelled relationships between ca 150,000 soil profiles and 158 remotely sensed covariates¹⁵ and is at a suitable resolution for LDN target setting. In addition, as the variation of SOC stocks in space is several times larger than the variation of

¹⁴ ISRIC – World Soil Information. <http://www.isric.org/>

¹⁵ MODIS land products, SRTM DEM derivatives, climatic images, global landforms and lithology maps (Hengl et al., 2016)

SOC stocks in time (e.g. Conant et al., 2011) it is considered a suitable source of information in the absence of national SOC stock estimates for the year 2000.

To obtain some estimate in the **change of SOC stocks** suitable for LDN target setting, a modified Tier 1 IPCC methodology for compiling National Greenhouse Gas Inventories is employed to predict SOC trends at country level based on land use/cover change (IPCC, 2006).

Therefore, the dynamic component informing SOC trends is the land use/cover information showing change from one land use to another. Such changes are combined with the SoilGrids estimate, the general bioclimatic zone and minimal assumptions about the level of agricultural inputs in the new land use/cover, in order to make coarse estimates of SOC stock change¹⁶ using change factors (e.g. see Table 3.3.4 in IPCC, (2006) for Cropland change factors). Changes in land use/cover are assumed to have occurred for a 10 year duration which is reasonable as land use/cover data represents epochs centered around 2000 and 2010. Additional to the IPCC methodology, where land cover has moved from a degraded type (e.g. bare land) to a less degraded type (e.g. forest or grassland) an inverse relationship is employed to estimate restoration of SOC stocks.¹⁷ The loss of SOC from soils previously vegetated upon conversion to bare lands is also estimated as a 90% loss in 20 years, intended to reflect both degradation and surface soil erosion (implied in IPCC 2006 text and implemented here).

At this time, it is not possible to obtain (global) information for management effects on SOC stock within land use/cover types. Therefore LPD are the best indicator for potential effects on SOC stocks for land remaining in the same land use/cover category. However, average SOC stocks have also been provided for land remaining in the same land use/cover category.

For the LDN target setting, in the absence of a national SOC database, it is recommended to use the SOC 0-30 cm stock product derived from SoilGrids250m as a stand-in for baseline 2000 SOC stock. The value of SOC provided in the data set is a continuous variable ranging from 0 (shifting sands, bare rock, ice, water) to 800+ tons (wetlands/peatlands in highlands and cold climate) per hectare¹⁸. Optional classification to fixed intervals of SOC contents could be considered but should be done separately for each country to represent the relevant and most meaningful sub-division for the environmental conditions of the respective country.

¹⁶ Currently available global spatial data sets are not adequately supported by temporal SOC measurements to be able to provide trends. ISRIC and the Global Soil Partnership (GSP) are currently elaborating options for future global measurement programmes that may allow the establishment of spatially distributed SOC trends. Countries with existing soil profile data that are seeking improved SOC predictions for their territories or other assistance can contact ISRIC directly (see <http://www.isric.org/> or tom.hengl@wur.nl). At regional levels (e.g. in Africa, Australia and Europe), initiatives exist that aim at the establishment of methodologies and protocols for regional scale SOC measurement. This type of initiatives could produce regular up-dates of spatially disaggregated SOC data both for wide areas, especially of agricultural land e.g. Aynekulu et al. (2011), Lugato et al. (2014); and for smaller, specific project/farm scale areas e.g. de Gruijter et al 2016.

¹⁷ Where the loss of the opposite change (e.g. Forest to Cropland) is assumed to be the potential amount of gain in SOC from switching land use/cover.

¹⁸ The June 2016 version of SoilGrids250m is known to over predict the SOC stocks for soils with higher than 8% SOC (~4% soil profiles) due to issues with the bulk density layer not decreasing quick enough with increasing SOC % (see <https://github.com/ISRICWorldSoil/SoilGrids250m/issues/27>). An updated version will be provided to countries as soon as available. For now, SOC stocks should be considered as an overestimate. Any other local issues (such as artifacts or unusual values) identified by national teams are encouraged to lodge a description here <https://github.com/ISRICWorldSoil/SoilGrids250m/issues> to improve the product. Local data can also be incorporated in subsequent updates upon request.

The general file specifications of country specific SOC data are:

- Coordinate system : WGS_1984 and projected MODIS sinusoidal (SR-ORG:6842)
- Cell size (X, Y): 0,0027777777, 0,0027777777 [decimal degrees] and cell size (X, Y): 300, 300 [meters]
- SOC values are continuous in metric ton per ha
- Change factor values are stored as integers (change factor x 10)

3.4. Data formats

Default data are provided as geo-referenced spatial layers both in raster (geo-tiff) and vector formats (shape files), both readable and usable with a broad range standard commercial and open-source GIS packages¹⁹.

National estimates of the indicators for the period 2000-2010 are also provided as numerical values. Figure 1 presents a snapshot of the tables which summarize the national basic default data for a country.

In Figure 1, the first table contains the area of each land use/cover category for 2000 and 2010, the net change in area for each, the net Land Productivity Dynamics (NetLPD) and the SOC stock baseline for the year 2000. For example; in row 6, forest areas account for 4274 sq km in 2000 and 3963 sq km in 2010, with a loss of 311 sq km during the 10 year interval. The NetLPD is provided for all land use/cover that remains unchanged between the periods. For instance, 13 sq km area of 'forest remaining forest' has land productivity showing 'early signs of decline'.

In the second and third tables, only the areas with a change in land use/cover category have been considered. For example, 74 sq km has changed from 'forest to cropland' and 6 sq km of this change is showing 'early signs of decline'. A total SOC loss of 40,518 tons has been estimated for the 'forest to cropland', representing ~0.01% of the national soil carbon stock.

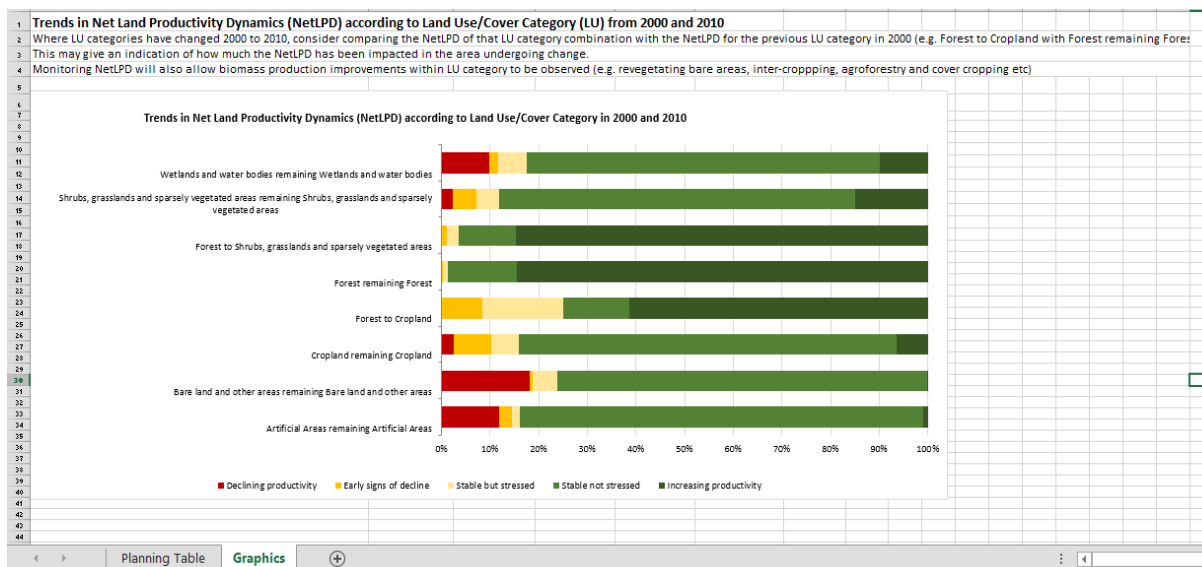
In Figure 2, the graph visually represents trends in Net LPD according to combinations of land use/cover in 2000 and 2010. This graph is intended to facilitate visual assessment of the intensity and persistence of negative or positive alterations of land productivity in changed and unchanged land use/cover categories. For example, only 1.3% of 'forest remaining forest' areas have NetLPD values of 1, 2 or 3 (indicating decreasing land productivity) representing ~ 53 sq km, whilst 25% of 'forest to cropland' areas show a trend of decreasing land productivity (~18 sq km).

¹⁹ Including ArcGIS, SAGA GIS, QGIS and R.

Figure 1: National basic default data

LDN Target Setting Programme										
Table 1 - Presentation of national basic data using the LDN indicators framework										
Land Use/Cover Category	Area (2000)	Area (2010)	Net area change (2000-2010)	Net land productivity dynamics (NetLPD)**						Soil organic carbon (2000)**
	sq km*	sq km	sq km	Declining	Early signs of decline	Stable but stressed	Stable not stressed	Increasing	No Data***	ton/ha
Forest	4274	3963	-311	1	13	39	564	3345	1	38.2
Shrubs, grasslands and sparsely vegetated areas	71341	71578	237	1667	3517	3211	52205	10665	75	25.8
Croplands	196808	196882	74	5131	14786	11468	152492	12750	181	18.5
Wetlands and water bodies	935	935	0	61	10	37	448	61	319	52.9
Artificial areas	328	328	0	39	9	5	272	3	0	18.4
Bare land and other areas	1417	1417	0	255	9	72	1069	1	12	15.0
SOC average (ton/ha)										20.8
Percent of total land area				3%	7%	5%	75%	10%	0%	
Total (sq km)	275104	275104	0	7155	18344	14832	207050	26825	588	
Changing Land Use/Cover Category	Net land productivity dynamics (NetLPD) trend 2000-2010									
	Declining	Early signs of decline	Stable but stressed	Stable not stressed	Increasing	Total				
Forest → Cropland	0	6	12	10	46	74				
Forest → Shrubs, grasslands and sparsely vegetated areas	0	2	6	28	200	237				
Changing Land Use/Cover Category	Net area change (2000-2010)	Soil organic carbon 0 - 30 cm (2000-2010)								
	sq km	2000 ton/ha	2010 ton/ha	2000 total (ton)	2010 total (ton)****	2000-2010 loss (ton)				
Forest → Cropland	74	23.3	17.8	172647	132129	-40518				
Forest → Shrubs, grasslands and sparsely vegetated areas	237	25.8	25.8	611784	611784	0				
Total	311			784431	743913	-40518				
Percent loss total SOC stock (country)						0.01%				

Figure 2. Graphical representation of Net Land Productivity Dynamics (NetLPD)



3.5. Validating default data

Should countries decide to use default data provided by the Programmes for LDN target setting, they are encouraged to validate them in view of their coherence with possibly comparable national data sets

Data derived from global or regional data sources should be contextualised with information at the national and sub-national level. The most common approach involves the use of site-based data to assess the consistency of the indicators derived from Earth observation and geo-spatial information. Another approach uses site-based data to calibrate and validate Earth observation indices and measures where the remote sensing variable is used to predict the same biophysical variable on the ground. The latter is also typically performed by data providers themselves as part of their product quality assurance procedure.

In the absence of site-based data, one cost-effective solution is to use multi-temporal high-resolution Google Earth images as verification data for tracking land cover patterns and their changes over time as verification of the default land cover data, but also to verify specific drivers of land productivity decline, which for example could be attributed in many cases to large uncontrolled urban sprawl or dynamic infrastructure developments like dam constructions and mine and quarry openings etc. User-friendly tools are being developed to facilitate this task (e.g. Collect Earth²⁰).

Qualitative information and stakeholder perspectives also play an essential role. A range of methods exist to gather these perspectives, including surveys, workshops, in-depth interviews, consultations and the establishment of expert panels. These methods are often based on the principles of expert elicitation (i.e. the synthesis of opinions of technical and scientific experts). Ultimately, it is likely to be the task of the lead organisation and its partners, who coordinate monitoring and reporting at national level in order to interpret what this combination of quantitative and qualitative data reveals about land degradation and restoration trends.

3.6. Using national data or alternative data sources

The global default data and tables provided for LDN target setting are populated using areas calculated from a global projection system²¹ rather than from any specific national projection system. Whilst the selected projection system (MODIS sinusoidal) is ideally suitable for calculating areas, it is not suitable for map visualization as 'true' country shapes are sacrificed for higher area accuracy. Additionally, should a country choose to calculate spatial extent of one or more LDN indicators in a different, national projection it would be expected that total areas would differ slightly, however proportional relations of differences (% change or composition of classes) will not vary.

If countries have well-maintained national datasets for the three indicators, they are encouraged to use them according to the following broad guidelines:

- Land cover: If a country's national land-use classification system does not match categories 1 to 6 as described above, the land-use classifications should be combined or disaggregated in order to represent the 6 broad categories presented here. Countries should report on the procedure adopted for the reallocation. The national definitions for

²⁰ <http://www.openforis.org/tools/collect-earth.html>

²¹ <http://spatialreference.org/ref/sr-org/6842/>

all categories used in the inventory and any threshold or parameter values used in the definitions should be specified. Countries will use their own definitions of these categories, which may or may not refer to internationally accepted definitions. Countries should describe and apply definitions consistently for the national land area over time.

- Land productivity: The breakdown of each land cover type according to the 5 degrees of productivity trend described in table 2 is recommended. To ensure some degree of comparability productivity figures, their trends and changes should be derived from time integrated land productivity information or proxies considering as much as possible full biomass growth cycles over a series of years and not just single date observations in separate years. Countries should describe data and methodology used, and should apply definitions consistently over time.
- Soil organic carbon stock: Should countries have their own national SOC data for baseline generation they are encouraged to either directly use this information or seek to have this information included in future updates of SoilGrids250m. Additionally, should countries have their own information on the trends involved in both land cover change and the effect of management within land cover classes (change factors for LUC, management within LU and/or inputs) they are encouraged to do so in accordance with Tier 2 methodologies for preparing National Greenhouse Gas Inventories (IPCC, 2006). Data must use standardised measurement units; i.e. tons SOC per ha for 0-30 cm depth.

4. Use of the indicators to assess LDN

The procedure for assessing LDN involves the setting the LDN baseline, assessing land degradation trends and monitoring progress towards LDN targets. While setting the baseline and assessing initial land degradation trends are to be conducted in the initial phases of LDN target setting; LDN monitoring is to be conducted at a point in the future (e.g. 2030, with potential intermediate monitoring points on an interval of 4 years) to track progress towards LDN targets.

4.1. Setting the LDN baseline

The **baseline is the initial numerical value of the three indicators** used as proxies of the land-based natural capital, at t_0 . The baseline values of these indicators will be re-measured at a point in future (t_1) to determine the change in land-based natural capital. The ambition of a country with respect to achieving LDN is no net loss, and thus **the LDN (minimum) target is equal to the baseline**.

To minimise the effects of seasonal and inter-annual climate variability the baseline value should be an average across an extended period (e.g. 10-15 years) prior to t_0 .

The exact identification of the baseline will ultimately depend upon available data series. For the default data made available by the Programmes, the baseline is *de facto* centered on the decade 2000-2010. Should countries decide to use national data, the baseline period might change. Countries should therefore clearly state the identified baseline period.

4.2. Assessing land degradation trends

Setting the LDN baseline is a stock-taking exercise where a snapshot of the current land-based natural capital is taken; it does not provide any information on the current status of land degradation.

An assessment of land degradation trends, coupled with an analysis of the driving forces behind these trends, is an essential step in terms of understanding current conditions of land degradation, revealing anomalies and identifying degraded areas. Such an assessment will provide an informed evidence base for setting sound LDN targets, making decisions about potential interventions and prioritising efforts in areas where degradation is taking place.²²

An observation of the changes in the value of the three LDN indicators over a 10-15 year assessment period can provide an indication of land degradation trends.²³

Whilst the three indicators together are suitable proxies for the ecosystem services provided by the land-based natural capital, there is no scientific basis for combining these into a composite indicator to give a single aggregated value. Aggregation would mask the changes detected in the individual measures, and would prevent the interpretation of individual measures at the national level based on local knowledge.

Positive change in one of the indicators cannot compensate for negative change in another because all are complementary, but not necessarily additive, components of land-based natural capital. Therefore, if one of the indicators shows a negative change, degradation is considered to occur, even if the others are positive.

In other words, this means applying a **‘one-out, all-out’ rule**. One-out, all out is the most comprehensive and conservative approach (following the precautionary principle) to using multiple indicators to assess status,²⁴ however recognizing that it is prone to false positive error.²⁵ More specifically, to apply the one-out, all-out rule each of the indicators needs to be enumerated and evaluated separately. Degradation is most likely to occur when:

- a) negative land cover change occurs; or
- b) NPP decreases significantly; or
- c) SOC decrease significantly.

However, **contextualising and interpreting changes in the indicators is likely to be the key task of the national and local authorities and institutions** that coordinate monitoring for national-level reporting. Relevant stakeholders should be involved in the interpretation of data.

Changes in land cover may be characterised as positive or negative when contextualised with national or local information. Some critical transitions are generally considered as negative, for instance those from natural and semi-natural land cover classes (e.g. forest,

²² For more information on how identify drivers of land degradation, set LDN national voluntary targets and associated measures see UNCCD, 2016. “Land Degradation Neutrality Target Setting. A technical guide. Draft for consultation during the Land Degradation Neutrality Target Setting Programme inception phase”.

²³ Many countries have already conducted assessments of the status and trends of land degradation using a variety of different approaches (e.g. Land Degradation Assessment in Drylands (LADA), carried out by the FAO) and implemented some form of land evaluation for the assessment of the land potential. In order to leverage the efforts made, countries are encouraged to use the results from those assessments to underpin the LDN target-setting process. In the absence of such assessments, or to complement existing information, countries are encouraged to apply the approach presented here.

²⁴ Source: Adapted from: European Communities. 2013. Overall Approach to the Classification of Ecological Status and Ecological Potential, Guidance Document No 13, European Union, Luxembourg.

²⁵ Borja, A, T. Prins, N. Simboura, J.H. Andersen, T. Berg, J.M. Neto, J. Reker, H. Teixeira and L. Uusitalo. 2014. Tales from a thousand and one ways to integrate marine ecosystem components when assessing the environmental status. *Frontiers in Marine Science* 1(Article 72):1-20. doi:10.3389/fmars.2014.00022

shrubs, grasslands and sparsely vegetated areas, wetlands) to cropland or to artificial areas; from natural and semi-natural land cover classes or cropland to artificial areas (i.e. urbanization); as well as from forest land to other land cover classes (i.e. deforestation).

Potentially negative trends are 12, 13, 15, 23, 25, 26, 35, 36, 43. Potentially positive trends are 51, 52, 53, 21, 31, 61, 62, 63. Land cover data can also be aggregated to estimate the following critical transitions:

- Conversion from natural and semi-natural ecosystems to agriculture:
 - Hectares of changes from forest, shrubland/grassland and wetland to cropland per square kilometer: $\frac{(13+23+43)}{(\text{total area of } 1+2+4 \text{ in } 2000)}$.
- Conversion of forest to other land categories (deforestation rate):
 - Hectares of changes from forest to shrubland/grassland, cropland, wetland, artificial areas, or bare land/other areas: $\frac{(12+13+14+15+16)}{(\text{total area of } 1 \text{ in } 2000)}$
- Conversion to artificial areas:
 - Hectares of changes from forest, shrubland/grassland, cropland and wetland to artificial area: $\frac{(15+25+35+45)}{\text{total area of forest, grassland, cropland, wetland } 2000}$.
- Conversion to bare land:
 - Hectares of changes from forest, shrubland/grassland, cropland, and wetland to bare soil: $\frac{(16+26+36+46)}{(\text{total area of } 1+2+3+4 \text{ in } 2000)}$.

Areas with increasing land productivity and SOC stocks can generally be interpreted as improving, while areas with **decreasing land productivity or SOC stocks** can be interpreted as deteriorating (i.e. degradation).

However, under exceptional circumstances, these indicators can lead to “false positives”. For example, bush and tree encroachment (i.e. land cover change from grassland to shrub-dominated) in the drylands often leads to a loss of natural capital with less forage for grazing animals and wildlife. It is therefore considered to be land degradation, even though the land productivity and SOC may both be increasing.

In the case of land productivity, it is recommended to pay particular attention in all land cover classes to the areas classified with the codes 1 to 3. Data can be aggregated to calculate:

- Rate of forest productivity decline, expressed in % of the total area of forest land: $\frac{(\text{forest area with codes } 1, 2, 3)}{\text{total forest area } 2010}$
- Rate of grass productivity decline, expressed in % of the total area of grass land: $\frac{(\text{grassland area with codes } 1, 2, 3)}{\text{total grassland area } 2010}$
- Rate of cropland productivity decline, expressed in % of the total area of crop land: $\frac{(\text{cropland area with codes } 1, 2, 3)}{\text{total cropland area } 2010}$

Encroaching grassland and cropland under intensive mono-cropping deserve special attention:

- Grassland encroachment is a direct consequence of overgrazing of grassland. It can occur in any grassland ecosystem. When grass is overgrazed, grass dies and bushes invade the space left bare. The land cover subsequently becomes a forest-like ecosystem, which is neutral or positive from the point of view of carbon storage. It is

possible to return progressively to grassland if sustainable grazing practices are applied but it takes resources and time (removing the bush after several years of soil restoration and re-establishing herbaceous species afterwards).

- In the case of intensive mono-cropping, biomass generally improves in the first years, thanks to input of chemical fertilizers. But soil carbon content declines. After a few years (10-20 years), yields are reduced (and therefore land productivity) due to loss of soil structure and alteration of dominant microbiological processes (and therefore loss of carbon content) and acidification.

Generally the areas of concern are well known by national ministries of agriculture and research centers, and can be easily identified on publicly available high-resolution satellite imagery servers such as Google Earth or the equivalent.

Numerical estimates of all indicators are provided to countries in excel tables to facilitate the identification of critical processes. In addition to the tables presented in Figure 1, which include estimates of changes in SOC stocks associated to land cover change, the graph shown in Figure 2 is intended to facilitate visual assessment of the intensity and persistence of negative or positive alterations of land productivity in changed and unchanged land use/cover categories.

4.3 Monitoring progress towards LDN

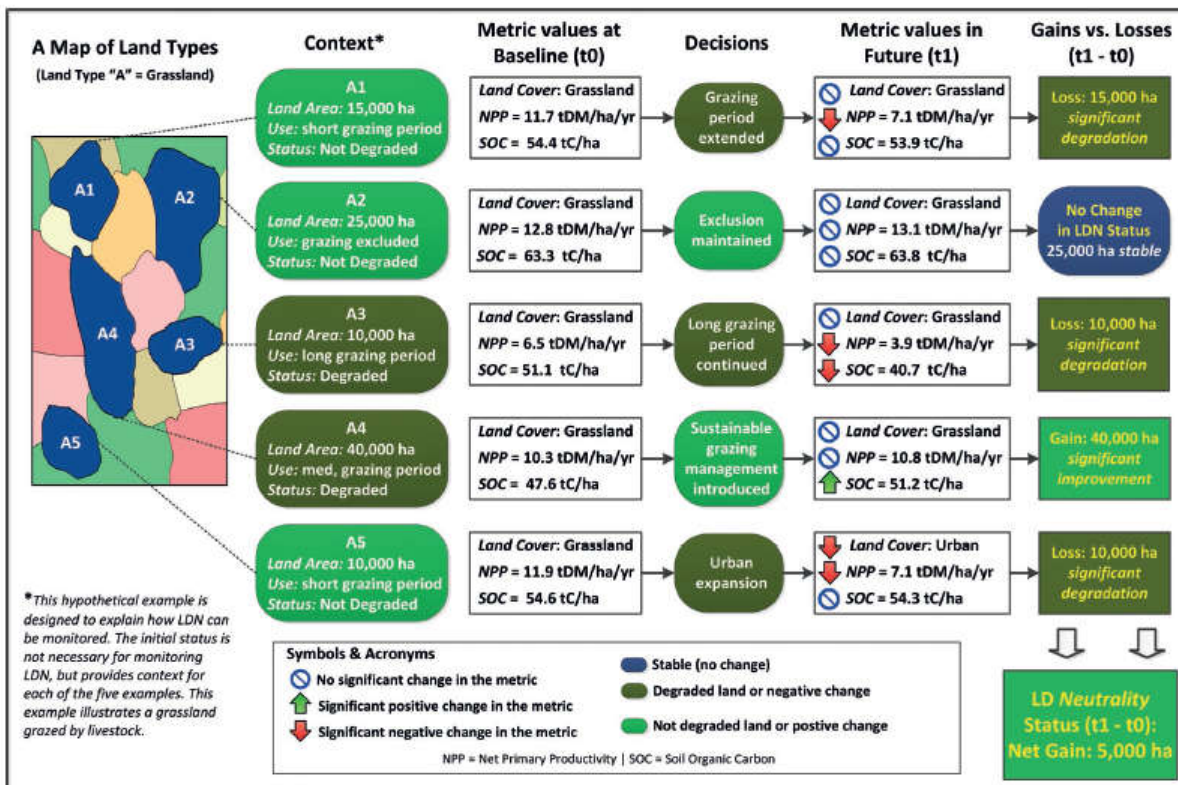
In order, to monitor progress towards LDN targets, each of the sub-indicators should be quantified again (using the same methods employed at baseline) at a point in the future, t1 (e.g., 2030, with potential intermediate monitoring points on an interval of 4 years).

The future monitoring points may comprise shorter periods than used to create the baseline (e.g., 5 years) to limit overlap with the baseline measurement period.

To determine significant changes, t0 and t1 values are compared. Monitoring neutrality involve both quantifying the area of significant new negative changes (losses, i.e. degradation) and significant new positive changes (gains) compared to the baseline. A loss is measured as a negative land cover change or a statistically significant decrease in land productivity or soil organic carbon stocks. A gain is measured as a positive land cover change or a statistically significant increase in land productivity or soil organic carbon stocks. Neutrality is achieved when the area of losses is equal or less than the area of gains, within each land cover class, and across land classes, at national scale. Figure 3 exemplifies the monitoring of LDN for a hypothetical land type.

The future UNCCD reporting process and possible linkages to the LDN indicators are expected to be decided by country Parties during COP 13 in September 2017.

Figure 3. Monitoring LDN based on changes in the value of the indicators, using the one-out, all-out approach (Source UNCCD/Science-Policy Interface, 2016).



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ANNEX I - Data accuracy and limitations

Land cover change

The adoption of a common land cover/use classification system implemented at global level ensures the harmonisation and standardisation of the land cover analysis and a certain degree of inter-comparability between countries:

- Advantages: same classification techniques and classification system used;
- Disadvantages: lower accuracy than using a specific land cover data source for each local ecosystem with higher spatial resolution.

Generally, a higher spatial resolution can allow improved accuracy as smaller differences between land cover can be distinguished²⁶. Indeed, for several countries there will likely be suitable regional datasets offering a relatively fine spatial resolution which can be considered as a good option as long as a country is equipped and qualified to handle this type of complex datasets.

However, a global map will never be better than a high quality national map. A high quality national map is produced using good quality input data (satellite imagery and aerial photos), a robust methodology following internationally accepted standards and a high level of quality control (ground truthing). Global land cover maps are done by processing satellite imagery mosaics with advanced and tested algorithms (automatic classification) while national land cover maps are usually done by supervised (or semi-automatic) classification with significant inputs of local analysts not only in the pre-processing phase (definition of training sets for the classifiers), but also in the post-processing for correction and post-editing. The country LDN monitoring and reporting should therefore be based principally on high quality national data whenever available.

The ESA Climate Change Initiative (CCI) Global Land Cover dataset (hereafter called ESA-CCI-GLC), is a high quality and reliable dataset which has undergone extensive global validation for each of the two epochs delivered, 2000 and 2010 (2005 is also available but is not used for the LDN target setting).

A critical step in acceptance of these LC maps by the wider user communities has been providing confidence in their quality through validation against independent data such as ground-based reference measurements and alternate estimates from other projects and sensors. Such a validation process was undertaken and ensured: (i) independent validation datasets were used (data that was not used during the production of the LC maps) and (ii) the process was carried out by external parties, (i.e. by staff not involved in the production of the LC maps)²⁷.

²⁶ However, sources of error still derive from the type of algorithms used, the manner of classification, differences in sensors/dates/epochs etc

²⁷ More information on the validation procedure and results of the ESA CCI land cover maps can be found at: <http://maps.elie.ucl.ac.be/CCI/viewer/download/ESACCI-LC-PUG-v2.5.pdf>

The validation included three different steps: elaborating a sampling strategy (usually a stratified random sampling according to the different land cover classes mapped, in order to have a representative reference data sets for each of the classes), collecting reference data sources and assessing the product accuracy. For global land cover products (such as the CCI-LC maps), collecting global “ground thruthing” (visiting many points across all land masses) is very costly. However, surrogate “ground thruthing” was obtained from existing reference data sources by expert interpretation. The following reference data sources were used by an international network of experts (covering different regions of world) in the validation process:

- High and very high spatial resolution imagery available from Google Earth/Virtual Earth imagery, plus 1 Landsat TM or ETM+ image over each epoch obtained from the Global Land Survey (GLS) dataset from 2000, 2005 and 2010;
- Multi-temporal Normalized Difference Vegetation Index (NDVI) profile derived from SPOT-VGT time series for the years 2000/2005/2010; and
- Google Earth facilities.

The overall accuracy values were weighted by the area percentages of various land cover classes (number of samples proportional to the surface of each land cover class). The weighted-area overall accuracy result of the CCI-LC map is between 73.2% and 74.1%, referring to the overall accuracy assessment for the 22 full classes (with weighted producer/user accuracies depending on the importance of the subject classes). A similar analysis for the 6 aggregated classes used in the LDN TSP would give a higher value. Indeed, even if some forest sub-classes might have low accuracy values (as some different types of forest are difficult to distinguish), overall the aggregated 'forest' class has higher accuracy (as the different types of forest difficult to distinguish are integrated in a simple non ambiguous class).

In the original ESA-CCI-GLC dataset, different land cover classes have different level of accuracy:

- “Bare soil areas”, “broadleaved evergreen forest”, “water” and “snow and ice” (the most unambiguous and spectrally homogeneous and recognisable classes) have high accuracy.
- The various cropland and forest classes have slightly lower relative accuracy as they are less easily distinguished but when sub-classes are aggregated, the accuracy of those two aggregated classes is notably high.
- Classes such as “shrub land, grassland, sparse vegetation” and “wetlands” were more readily affected by misclassification errors. This is due to the mixed or composite nature (as opposed to homogeneous) of this type of vegetation cover.

Regarding the percentage of the accuracy performance, the correct interpretation is that the total accuracy of the validated areas, according to the sampling design, indicates the area percentage correctly classified by the automatic classification algorithm used (higher accuracy means that the algorithm is more accurate). The remaining validation areas were predominantly unclearly classified (where two different but close classes return a similar probability, although the “close classes” were in most cases aggregated within the same class in the aggregated legend used in the LDN TSP, ensuring higher accuracy) and small proportions were wrongly classified.

About the Land Cover change assessment, one major source of classification uncertainty in the ESA-CCI-GLC is that the two epochs released have been built on satellite images covering a time span of 5 years: 2000 (1998-2002) and 2010 (2008-2012). This means that if there were some LC changes during these time spans, the classification will attribute the most dominant class during the period, and the changes will not be detected.

Another point which could create uncertainty in the land cover change detection is the seasonality. Use of images from different seasons (e.g. the dry season vs the wet season) can sometimes produce controversial interpretations due to drastic changes in vegetation cover at different times of the year.

Seasonal factors are most marked in:

- Natural vegetation areas: e.g. woody vegetation in the dry season is leafless and its total cover can appear less extensive than it is.
- Wetland and water areas: according to the land cover classes present in the legend which detect perennial e not perennial natural water body standing or flowing, it isn't possible to classify the differences due to seasonal factors or link to the depth of a water body itself.

The processing chain of the ESA-CCI-GLC is designed to capture the seasonality (inter-annual temporal variation) and hence should not be much affected by the acquisitions times of images from different seasons, unless these images were not available due to atmospheric perturbations (such as clouds). The Land Cover CCI used as input 7-day composites images and the seasonality of the land cover was an essential input to the classification system for distinguishing classes that are similar spectrally but that have a totally different temporal behavior.

Although the ESA-CCI-LC dataset was not made to provide land cover changes *per se*, the comparison of the two epochs (where data is aggregated over several years to remove the effect of seasonality and to meet the need to collect enough images in cloudy climate condition areas where one year was not sufficient to acquire images with no cloud cover for an acceptable classification) generally allows for the identification of major (aggregated) land conversion trends – sufficient for LDN target setting for all classes, except urban and to a lesser extend wetlands. The dataset shows some shortfalls in the urban and wetland layers, meaning the changes related to those 2 classes are underestimated. But this must not delay LDN target setting. These classes will be addressed in the next release of the ESA-CCI-GLC²⁸ that will be soon available.

The new release (expected in January 2017) will have the LC changes as an “integral part” of the processing chain and consequently accuracy of land cover changes will be higher, especially for urban and wetlands, because land cover maps more consistent in time.

²⁸ To improve the urban layer, the next release of the ESA-CCI-GLC dataset will integrate in its processing chain the JRC's Global Human Settlement Layer (<http://ghsl.jrc.ec.europa.eu/>). For the wetland layer, some improvements are expected as a result of a number of wetland inventory activities conducted under the auspices of the recently formed GEO Wetlands initiative (e.g. ESA's GlobWetland Africa and EC H2020 SWOS) but the availability of an improved wetland layer in the ESA-CCI Land Cover dataset will require more time.

Land Productivity Dynamics

The 5 classes of the Land Productivity Dynamics data set (LPD) integrate over a 15 years observation period from 1999 to 2013 information on the direction, intensity and persistence of the trend and change of above-ground biomass generated by photosynthetically active vegetation cover, widely equivalent to Gross Primary Production (GPP) of the global land surface.

Within one pixel (for LPD 1 km²) low-resolution imagery may typically assemble a considerable amount of vegetation heterogeneity, and above-ground biomass production is not to be equated with crop production. Consequently, it must be clearly understood and communicated that 'land productivity' in the context of the LPD dataset strictly refers to the overall above-ground vegetation biomass productivity. This is not conceptually the same as, nor necessarily directly related to, agricultural income per area unit or 'land productivity' as used in conventional agricultural terminology. Furthermore it has to be understood that the 5 LPD classes provided are not associated to specific levels of above-ground biomass production or specific biomass quantities lost or gained during the observation period. Each class characterises primarily the overall direction, relative change intensity and persistence of gross primary production independently of the actual level of vegetation abundance or land cover type. This means each LPD class can appear in any type of land cover and any level of vegetation density. Nevertheless, the quantitative information on biomass productivity levels is contained in the input NDVI time series data. From this, one could, for example, extract the average annually integrated NDVI over a reference period of e.g. 3 to 5 years as baseline GPP proxy as well as subsequently the % deviation (positive or negative) from the baseline in defined time steps (e.g. every 5 to 10 years) for the monitoring of progress towards LDN targets.

Input data used for the current LPD product is an operational global 15-year time series of daily SPOT-VGT NDVI images aggregated/composited to 1 observation every 10 days (i.e. 540 observations overall for each pixel). As other operational global remote sensing time series products, typically generate by national and international space agencies following comparable standards, this time series includes corrections for radiometric system specifications, atmospheric effects, illumination conditions and cloud detection to obtain continuous standardized vegetation indices over time. Subsequently it has undergone a very detailed validation process which was carried out by comparison with other operational NDVI time series and against global field validation/calibration networks. Overall, no significant differences were found between the LPD input dataset and other operational validated NDVI time series over the same geographic areas²⁹. This is an important pre-requisite assuring that the LPD processing chain can be applied independently from a specific input product with comparable results. Despite the operational pre-processing, all available global NDVI time series products contain no data pixels due to correction artefacts, which often concentrate over areas affected by regular formation of convective clouds (e.g. small islands, tropical rain forest, high mountain areas), as well as areas with permanent ice and snow (high mountains) and large ephemeral water bodies. Consequently, these conditions often also lead to no data pixels in the LPD classification. Nevertheless, these LPD no data pixels can be usually explained by the land cover context in which they are found and can then be accordingly accounted for in the disaggregation by land cover in the LDN planning table. For example, "no

²⁹ Validation Report SPO-VGT (GEOV1)

http://land.copernicus.eu/global/sites/default/files/products/GIOGL1_VR_NDVI-VCI-VPI_I1.21.pdf

data” in wetlands is typically associated with open water bodies counted in the total LC wetlands area. Also in high mountain areas “no data” pixels typically fall in the bare land class due to frequent cloud cover and sometimes ice and snow covers.

Validation of LPD classes is not a trivial task because typically no directly comparable field data on land productivity change is available. Nevertheless, validation of LPD classes in terms of plausibility testing against the land cover change detected by the CCI LC data set and locally against multi-temporal high resolution data in Google Earth has been performed. A preliminary global statistical validation of LPD classes can be and was performed against mapped land cover changes between the CCI LC epochs 2000 and 2010 considering the full range of mapped CCI LC classes and not only the 6 aggregated IPCC classes. The area of mapped LC change globally covers approximately 278061 km² accounting only for about 2.44 per mill of the vegetated land surface (see Figure 1 below)

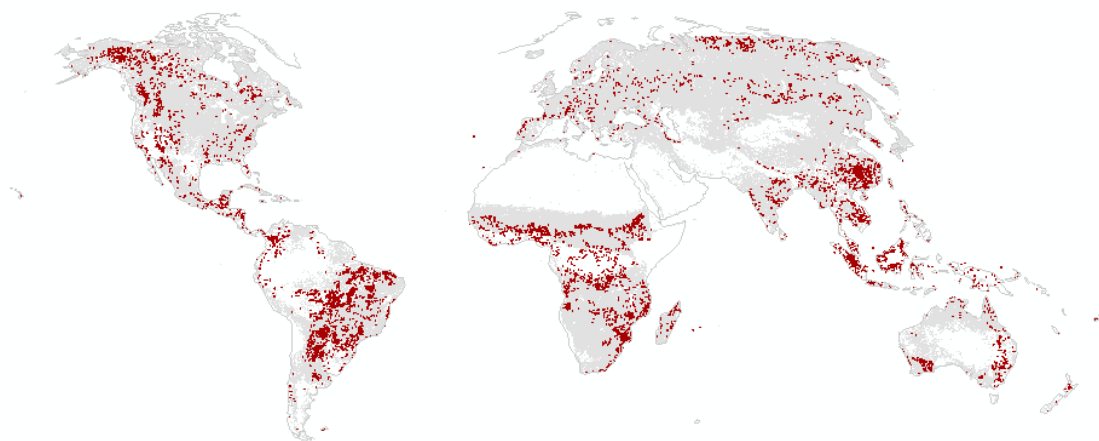


Figure 1: Areas with land cover change between 2000 and 2010. Area extents are exaggerated in order to be visible at presented scale.

For a number of critical LC transitions, cross correlation between the expected LPD class distribution in relation to observed land cover changes were investigated and further verification is on-going. For example transitions from semi-natural LC classes with tree cover to bare and sparsely vegetated areas are expected to feature predominantly LPD classes 1 to 3, while LPD 4 and 5 should be reduced, thus showing a clearly different picture than the general global LPD class distribution where classes 4 and 5 are the vast majority accounting for roughly 80% of all pixels.

In the case shown in Figure 2 a high level of correspondence between declining land productivity and independently mapped loss of vegetation cover expressed as land cover class change provides evidence of the plausibility and relative accuracy of the LPD class distribution. For other land cover transitions the correlation is less clear at global level (e.g. conversion from evergreen broadleaf forest to cropland) but initial steps to refined verification at regional to national levels indicate clearer and more plausible relationships between LPD classes and mapped LC change. Results of this refined validation will be made available as soon as possible.

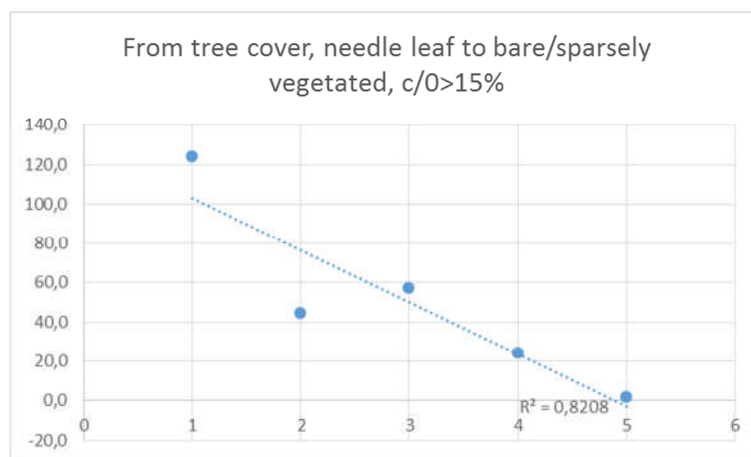


Figure 2: Distribution of LPD classes within areas changed from tree cover to bare/sparsely vegetated

However, the vast majority of LPD classes indicating a clear and persistent change of land productivity fall into areas where in the base data sets made available no mapped information of land cover change is appearing. Therefore, local verification using Google Earth multi-temporal high-resolution images is recommended as a quick option for searching verification of land productivity changes. The LPD geo-tiff class images can be easily up-loaded on Google Earth and interactively investigated against changes visible in the underlying high-resolution image data base. During the first LDN pilot phase 2014/2015 it could be shown in many cases that declining productivity classes in the LPD data were associated to urban and infrastructure expansion (e.g. dam construction, mine openings etc.) as driver of land productivity losses affecting ecosystem functioning in their wider surroundings.

However, the analysis of temporal efficiency levels of vegetation to obtain an indication of long term changes in that efficiency (the LPD product) is only a first input requiring other information for correct local/regional interpretation in a land degradation context. The LPD results must as much as possible be further integrated and contextualized with more additional information that reflects climatic and/or socio-economic factors such as local land use, changes in land use practices and/or yield outputs, population changes, etc. This integrative analysis is needed to obtain a holistic interpretation of possible ongoing land degradation that explains the biophysical dynamics in relation to anthropogenic drivers.

To test this integration, a first analysis at global level dealt with the correlation of some spots of decreasing productive capacity in the long time-series product (i.e., 15 years, from 1999 to 2013) against actual global drought monitoring data. These revealed strong correlations with areas having undergone recent and recurrent droughts.³⁰

Therefore, during the interpretation of the LPD dataset, although signs of declining and increasing land productivity will be found, it does not necessarily indicate conditions contributing to land-degradation neutrality. To establish the contribution of increasing and decreasing productivity of areas, an assessment by contextual analysis in relation to the major land degradation issues is required.^{31,32}

³⁰ CHERLET, M *et al*, 2015, Use of the NDVI to Assess Land Degradation at Multiple Scales

<http://link.springer.com/book/10.1007%2F978-3-319-24112-8>

³¹ EC-JRC, 2013, Land-Productivity Dynamics Towards integrated assessment of land degradation at global scales

<http://publications.jrc.ec.europa.eu/repository/bitstream/JRC80541/lb-na-26052-en-n%20.pdf>

³² EC-JRC 2013, on Land Productivity Dynamics in Europe

<http://publications.jrc.ec.europa.eu/repository/bitstream/JRC80540/lb-na-26500-en-n%20.pdf>

Soil Organic Carbon

SOC stock baseline

The soil organic carbon (SOC) stock maps for 0 – 30cm were created from a combination of three soil predictions from SoilGrids250m²⁸ – SOC%, bulk density and gravel content. As a result, the accuracy of the current product is a function of the accuracies of each of its inputs. The amount of variation explained for each map in a global 10-fold cross-validation was 64%, 76% and 56% respectively (see Figure 3). The amount of variation explained (or R2 value) for a continuous variable (like SOC%) can be understood as: 64% of the spatial variation in cross-validation (test) points being explained by the ensemble of models used. That is – the pattern of the SOC% map captures 64% of the information in the points. Information that is lost tends to be from smoothing of the local variation (see middle and bottom panel, Figure 4).

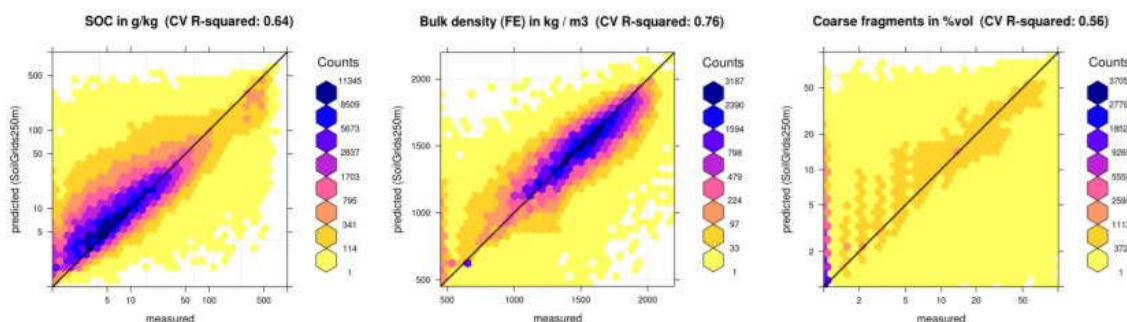


Figure 3: amount of variation explained for SOC%, bulk density and gravel content (coarse fragments)³³

As these accuracies are not a direct test of the SOC stock map itself and, due to recently found issues in the bulk density map (where soil bulk density values were not decreasing fast enough where SOC % is higher) a new SOC stock map is being prepared based on a direct prediction of SOC stocks. This layer will be supplied to countries once it is available. In the meantime, current estimates ought to be considered as an over-estimation where SOC% is above 8%. The proportion of SOC stock lost based on LC change however, will remain the same and so can be used for LDN target setting.

Limitations to the current maps used to construct the SOC stock primarily stem from predictions being based on soil legacy data and include:

- Measurements for SOC%, bulk density, gravel content and soil depth have been collected with different measurement methods (e.g. different laboratory methods, even when corrected for, introduce small amounts of noise)
- Soil data was collected over a large space of time (approximately 60 years for SOC, with the bulk centered on ~1995) and predictions were not made for the year 2000 (but assumed so in the absence of other, more suitable global data)
- Soil data was compiled from multiple sampling campaigns which were selected for their own purpose, that is – soil observations were not collected specifically for generation of SoilGrids, meaning that there may be sampling bias (e.g. an over-representation of agricultural areas is common) – see Figure 5.

³³ <http://www.isric.org/content/soilgrids> and Hengl, T., de Jesus, J. M., Heuvelink, G. B., Ruiperez, M., Gonzalez, M. K., Blagotic, A., ... & Guevara, M. A. (2016). SoilGrids250m: Global Gridded Soil Information Based on Machine Learning. *PLOS One*, in review. Available at: http://gsif.isric.org/lib/exe/fetch.php?media=wiki:soilgrids250m_global_gridded_preprint.pdf

- The collection of legacy data (WOSIS) is nowhere near exhaustive - much currently unaccessed legacy data exists in the databases of a multitude of agencies/companies in a multitude of languages.

Further details of its production, code and maps can be found at the ISRIC website.²⁹

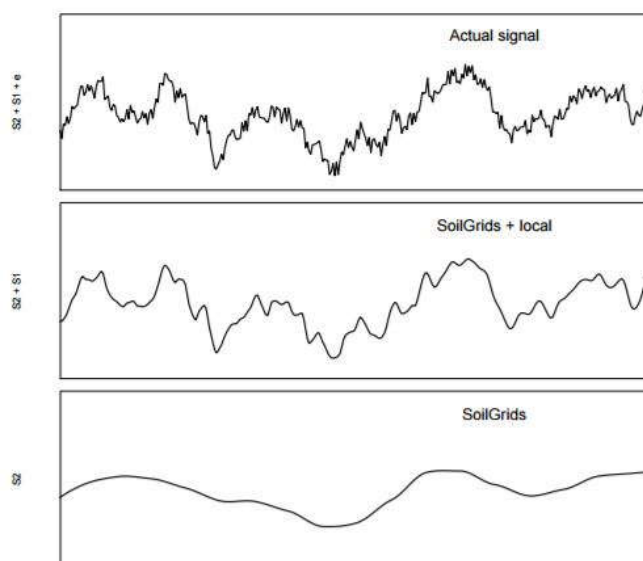


Figure 4: the amount of spatial variation captured by SoilGrids250m (bottom) as compared to SoilGrids plus local information (middle) and the actual signal in reality (top)³⁴

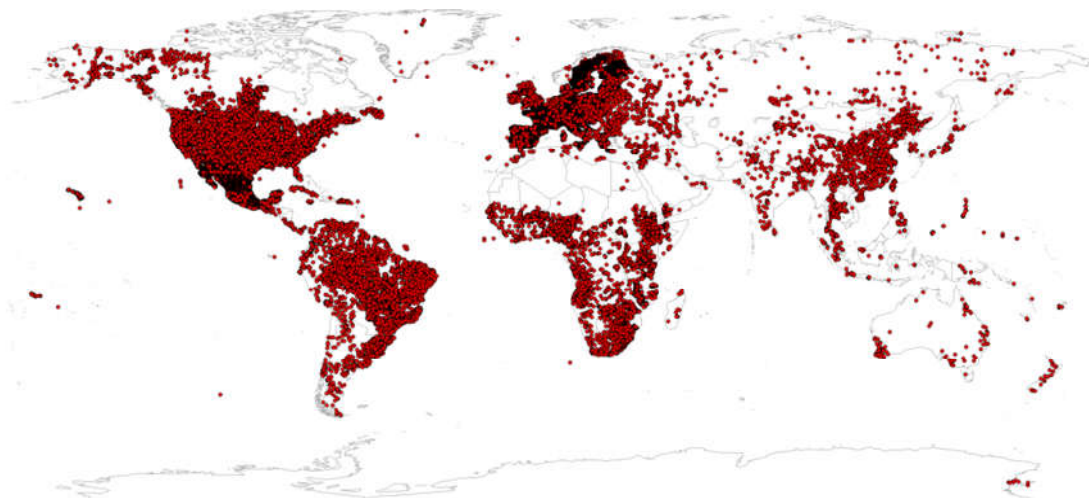


Figure 5: location of soil profiles included in the production of SoilGrids250m. Note large areas of clustering and large areas with few points⁸. To improve SoilGrids250m for your country visit <http://www.isric.org/content/faq-soilgrids>

³⁴ <http://www.isric.org/content/faq-soilgrids>

Where countries have their own soil profile data they may wish to have it included in WOSIS so that the SoilGrids maps will be improved for their areas. Alternatively, countries may wish to do their own mapping projects and in this case SoilGrids would be a useful covariate layer as it carries many relationships related to the soil forming factors for similar climate/elevation/LC etc.

In terms of validation of the supplied maps, it is recommended that countries validate against actual observations of SOC (if they have them) and not soil polygon maps. However, this process is not considered necessary for LDN target setting.

Change in SOC stocks

Changes in SOC stocks are based on change factors modified from the Tier 1 IPCC methodology for compiling National Greenhouse Gas Inventories and are employed to predict SOC trends at country level based on land use/cover change³⁵ (e.g. see Table 3.3.4 in IPCC, (2006) for Cropland change factors). As these change factors are based on collations of field trials and long-term experiments, they themselves come with confidence measures that may be applied by countries to their estimates of change to determine if they are significant¹⁰. The accuracy of change factors can be considered as the average case of response for that LC change for a given climate.

Limitations of this method include the lack of change factors for some climates (the nearest was used) as well as a paucity of change factors that could be applied to positive trends – that is, for LC changes where SOC stocks could be expected to increase. Additionally, only rough assumptions could be made as to inputs after LC change and no assumptions were made as to management of land after undergoing change. Should countries have their own information on the trends involved in both land cover change and the effect of management within land cover classes (change factors for LUC, management within LU and/or inputs) they are encouraged to do so in accordance with Tier 2 methodologies for preparing National Greenhouse Gas Inventories (IPCC, 2006). Data must use standardised measurement units; i.e. tons SOC per ha for 0-30 cm depth.

For LDN target setting

Where countries use the default data for target setting the following local information may be considered:

- Does the management of land after LC change tend to decrease SOC stocks (and so what management interventions in the new LC could be set as targets to alleviate said loss)
- After examination of local information, are SOC losses associated with soil loss? (and so what targets could be set to stabilize soils and prevent SOC loss from surface erosion)
- For other areas remaining under the same LC - how does the LPD align with areas that may have undergone SOC loss? Where LPD is declining, there is a greater chance of SOC loss occurring.
- Other, only locally understood information.

³⁵ IPCC (2006). Guidelines for National Greenhouse Gas Inventories. Volume 4: Agriculture, Forestry and Other Land Use. Geneva. Available at: <<http://www.ipcc-nggip.iges.or.jp/public/2006gl/vol4.html>>

ANNEX II - LDN Watershed Level Analysis

Summary of watershed analysis workflow:

- Watershed boundaries derived from FAO Hydrosheds are clipped with country boundaries derived from Global Administrative Unit Layers (GAUL) from FAO GeoNetwork.
- Potential priority watersheds are ranked by overlapping watersheds with the land use/cover (LUC), land productivity dynamics (LPD) and soil organic carbon (SOC) maps for the identification of areas where significant negative changes related to the land degradation neutrality (LDN) indicators occur (i.e. potential LDN hotspots).
- Based on the ranking, default potential priority watersheds are identified.
- Layouts maps and tabular data are shared with the countries for their feedback on the default watersheds and final selection on the priority watersheds (it is recommended that a maximum of 2 to 5 watersheds be selected and that selected watersheds be larger than 20.000 ha).
- Based on the feedback from the countries, data for the selected watersheds are made available by the LDN Target Setting Programme.
- An in depth analysis of the selected watersheds is conducted through the combination of a 30m Digital Elevation Model (DEM) with the main land cover types and land productivity values, to represent the LUC/LPD for a specific slope range.
- The watershed data package will include: the layouts maps at national level, the tabular data showing the ranked priority watersheds, the 'zoom in' maps of selected watersheds (i.e. LDN hotspots) representing a *slope* analysis approach.

1. Introduction

Contextualisation of LDN data layers at the watershed level is intended to:

1. Facilitate identification of priority watersheds and land degradation LDN hotspots; and
2. Provide decision support for the formulation of meaningful land use/restoration policy and LDN transformative projects/programmes at the country level.

It is hoped that the contextualisation of the three LDN indicators (LUC, LPD and SOC) with additional data will enable creation of targeted policies and support transformative action on the ground. The watershed level data will be used to provide countries with the additional context given by:

1. Watershed boundaries, stream networks, towns;
2. Landscape slope ranges;
3. Combinations of LDN indicators and slope classes.

The identification of the default potential priority watersheds, containing the LDN hotspots of degradation, is performed on the basis of the 3 LDN indicators and could provide a basis to support countries in cross checking with their own watershed priority and LDN hotspot lists.

The watershed level was selected as it offers a meaningful landscape unit framework, which will be, in many cases, a suitable basis for more detailed analysis for LDN hotspots identification and follow-up actions. The data contextualisation is accompanied by a proposal /example of how to meaningfully translate such a contextualisation in terms of land use/restoration policy and transformative projects/programmes.

2. Identification of priority watersheds

The main criteria applied for the identification of the default potential priority watersheds considers the 3 LDN indicators (i.e. LUC, LPD and SOC).

The watershed level analysis is conducted in two stages:

Stage 1: Country-wide watershed based analysis:

- Collation of LDN indicator combinations per watershed per country;
- Combination of indicators into a single factor representing 'area of degradation';
- Ranking and identification of default watersheds according to the degraded area percentage. From 2 to 5 default priority watersheds are identified and proposed for a deeper analysis, the number is chosen per country to maximise the coverage of degraded land for the smallest national area. The number of selected watersheds may vary from country to country according to the LDN indicators distribution.
- Feedback from countries on the default watersheds and final selection on the priority watersheds is welcomed. It is recommended that a maximum of 2 to 5 watersheds be selected and that selected watersheds be larger than 20.000 ha.

Stage 2: In depth analysis of selected watersheds with higher percentages of degraded areas.

- Examination of selected watersheds;
- Analysis of SOC thresholds per land use (Forest, Cropland, Grassland)
- Analysis of land use (Forest, Cropland, Grassland) + slope classes;
- Analysis of LPD + land use and slope classes;

2.1 Stage 1: Country-wide watershed based analysis

In order to allow the identification of default watersheds containing possible priority LDN hotspots, the collation of the following indicators for each watershed within a country is conducted:

- a. Area with declining LPD (LPD classes 1-3, i.e. Declining productivity; Early signs of decline; Stable, but stressed) (sq km)
- b. SOC loss (tons)
- c. Area of SOC loss (sq km)

The LPD dataset can indicate persistent levels of land productivity change or stability which reflect the dynamics of land transformations potentially impacting on land quality and is therefore used as the first step in the determination of areas affected by persistent and ongoing land degradation in an assessment. Consequently, areas indicated with declining land productivity LPD classes 1-3 may be first candidates to be considered for measures to halt land degradation.

In line with this it should be also understood that:

- Clearly declining land productivity i.e. LPD classes 1 and 2 in the data set point to areas that have lost land productivity between 1999 and 2013, this means a high probability of recently active land degradation processes.
- Stability but indications of 'stressed situation', i.e. LPD class 3, often represents strong inter-annual variations in land productivity, which could be an indication of beginning instability in land conditions.

SOC loss is also taken to represent the land use/cover change indicator, as SOC changes are computed based on LUC changes. The SOC loss indicator is then expressed in terms of area (sq km) in order to unify the three indicators on a common, unbiased basis.

The SOC area, including the negative land use changes, and LPD indicators are then combined using a one-out all-out approach in order to unify all 3 indicators into a single variable – **area of degradation (sq. km)**. The area of degradation per watershed is then used to rank watersheds in order of priority (see Table 1).

Priority	Watersheds	Degrading area: LPD + SOC (sq km)*	Cumulative sum degraded area (%)	Watershed area (sq km)	Cumulative sum national area (%)	SOC loss (tons)	Area SOC loss (sq km)**	Cumulative sum national SOC loss (%)	LPD 1-3 area (sq km)***	Cumulative sum LPD 1-3 (%)	Priority
	☒ Southern Central America	4350		51281		-189198	49		4301		
	General	611	14%	2531	5%	-2376	0.8	1%	610	14%	
	Coto Brus / Cot?n	338	22%	1619	8%	-6201	1.7	5%	337	22%	
	T?rcoles / Virilla / Colorado	312	29%	2144	12%	-28080	7.4	19%	305	29%	
	Terraba	294	36%	884	14%	-1260	0.5	20%	294	36%	
	Tempisque	243	41%	3374	21%	0	0.0	20%	243	42%	
	Sierpe	198	46%	2292	25%	-369	0.1	20%	198	46%	
	Guacimal / Sardinal / Seco	185	50%	956	27%	-5454	2.3	23%	182	50%	
	Bongo	169	54%	1185	29%	0	0.0	23%	169	54%	
	Parismina	165	58%	2852	35%	-16965	3.9	32%	161	58%	
	San Carlos	131	61%	3386	41%	-17028	4.2	41%	127	61%	
	Savegre	130	64%	1940	45%	-12051	3.5	47%	127	64%	
	Morote	119	67%	826	47%	0	0.0	47%	119	67%	
	Coto Colorado	116	69%	1446	50%	-1404	0.5	48%	115	69%	
	Tenorio	105	72%	629	51%	0	0.0	48%	105	72%	
	Nosara	102	74%	799	52%	-288	0.2	48%	102	74%	
	Fr?o	92	76%	1611	56%	-5427	2.4	51%	89	76%	
	Upper Lake Nicaragua	73	78%	707	57%	0	0.0	51%	73	78%	
	Teribe / Culubre	69	79%	242	57%	0	0.0	51%	69	80%	
	Pizote	66	81%	987	59%	0	0.0	51%	66	81%	
	Bebedero	65	82%	784	61%	-648	0.3	52%	65	83%	
	Barranca	62	84%	457	62%	-333	0.1	52%	62	84%	
	Chirrip?	61	85%	1688	65%	-9558	2.7	57%	58	85%	
	Saraoiqui	58	86%	3087	71%	-19467	4.3	67%	53	87%	

Table 1: Default ranking of watersheds in order of priority based on the area of degradation per watershed. In this example, the red priority level indicates the five default watersheds, covering 41% of degraded area and 21% of national area.

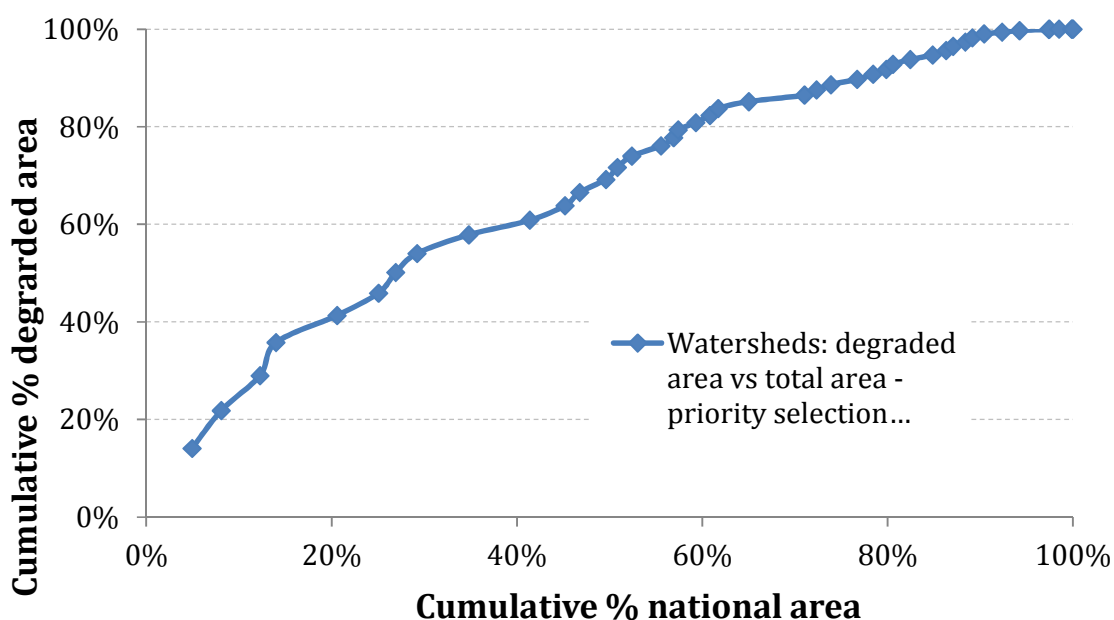


Figure 1: A comparison of ranked watersheds by their % total degraded area vs % total national area (as contributed by each watershed). These curves assist in assigning watersheds to the default priority levels for each country.

Thresholds between each priority level, whose priority is decreasing moving through the red, orange, yellow and grey colour, can be decided per country based on other variables such as the cumulative % of degradation addressed, cumulative % of national area, % area of LPD 1-3 or % area/tons of SOC decline. These priority levels can also be decided based on local data. The default analysis is only undertaken on data available at the global level. In the above example, we have considered the second and third level of priority (orange & yellow) to represent the area of degradation at which there qualitatively appears to be a diminishing return in terms of area of degradation addressed as per the size of each additional catchment (see Table 1 and Figure 1).

It is important to note that as the areas of SOC decline tend to be smaller than that of LPD 1-3, hence watersheds with higher SOC decline but low amounts of LPD 1-3 will occur lower down in the ranking. Therefore, countries could adjust their selection of priority watersheds according to internal considerations.

Country-wide combinations of indicators are intended to support the country based selection of priority watersheds. These include:

- LUC changes per watershed (see Figure 2)
- LPD 1-3 declining zones per watersheds (see Figure 2) and
- SOC functional thresholds per watershed (see Table 2)

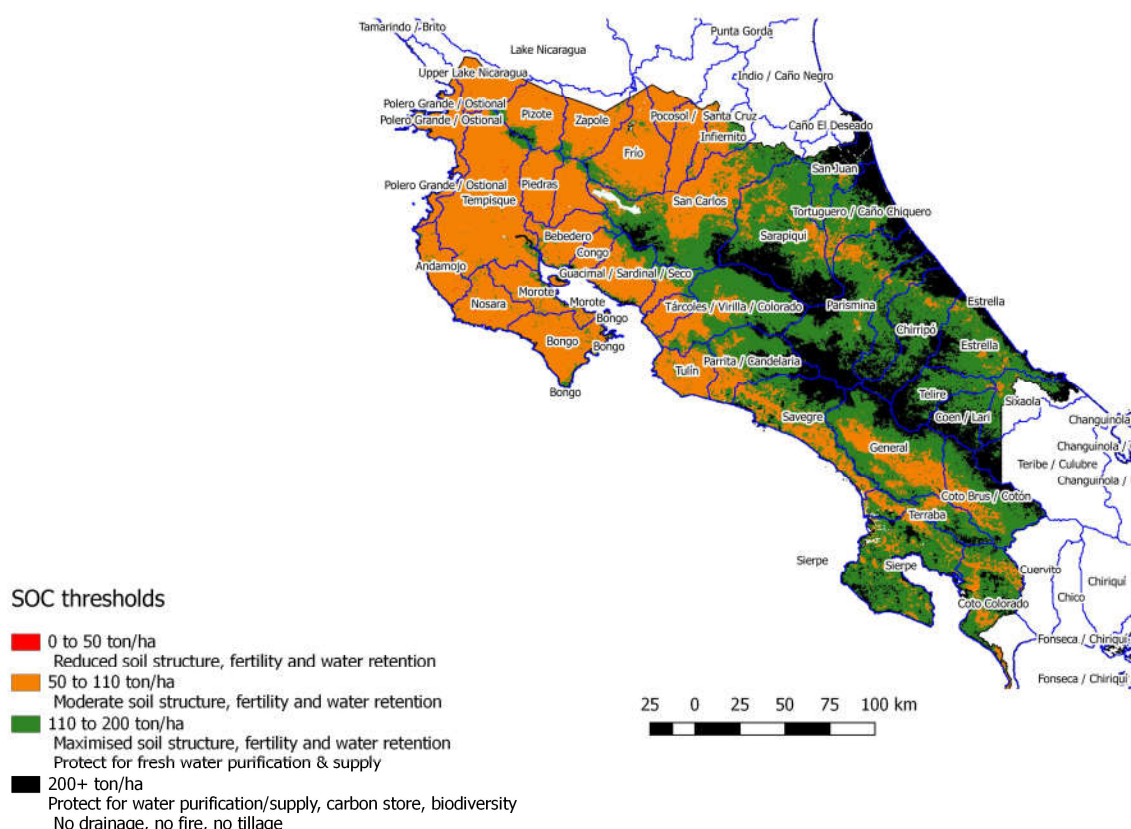


Figure 3: Generic soil functional thresholds as per those derived from the SOC level. Thresholds are accompanied by generic characteristics and some primary considerations.

To create country-wide combinations of LDN indicators, the watershed boundaries obtained from FAO Geonetwork³⁰ have been intersected with the country GAUL³¹ boundary. National maps were then compiled for the three indicators (LUC, LPD and SOC), derived from the default datasets provided to the countries, in combination with the watershed sub-basin boundaries. Each of these combinations was considered in relation to the default priority ranking to ensure that the default selection is meaningful.

The suggested criteria to be applied by the countries for the final selection of priority watersheds are:

- i) Minimum 20.000 ha area (200 sq km);
- ii) LPD showing a significant negative trend;
- iii) SOC showing a significant negative trend based on land cover change;
- iv) From a minimum of 2 to a maximum of 5 priority watersheds should be considered for the 'in depth' analysis.

³⁰ <http://www.fao.org/geonetwork/srv/en/main.home>

³¹ <http://www.fao.org/geonetwork/srv/en/metadata.show?currTab=simple&id=12691>

2.2 Stage 2: In depth analysis of selected watersheds with higher percentages of degraded areas

A more detailed analysis is computed on the selected priority watersheds is conducted in an attempt to tease out further information from the indicators in relation to each other (LPD 1-3 & LUC & change) and other contextual site information (slope, rivers, towns etc).

This deeper analysis takes in consideration the 'slope %' as a further element, which can be considered as a supporting *factor* or an additional *filter* for land use planning. The approach could be used by countries to identify areas of local degradation (LDN hotspots) within the selected watersheds.

While the slope can be an important supporting factor for the formulation of adequate measures to achieve LDN and is widely meaningful especially where water erosion and associated risks are an issue of major concern for the country, it should be noted that other land degradation processes and factors could be more appropriate in areas which suffer wind erosion, sand encroachment, salinization, overgrazing in flat terrain etc.

Selected watersheds are analysed through the combination of a 30m Digital Elevation Model (DEM) and the main land cover types where negative changes have occurred: cropland, forest and grassland. The combination of slope and main land cover types, can help discriminating areas of potential degradation according to the land use occurred within a specific slope range.

Slope ranges considered are:

- < 5%
- 5% – 15%
- 15% +

Degradation generally occurs where inappropriate land use occurs: e.g. cropland or grassland in high sloping area (slope > 15%) indicates a land use that will on average, quickly lead to degradation. Specific maps with a 'traffic light' colour range approach are produced to represent the possible combination of land cover types and slope % (e.g. Costa Rica, Figure 4).

Additionally, the SOC thresholds applied in Step 1 are further broken down by LUC sectoral level of forest, agriculture (croplands) and livestock (grassland) (e.g. Costa Rica, Figure 5). This analysis is provided at the country level, including the selected watersheds, and are accompanied by primary risks and general guidance per LUC. This brief acknowledgement of risks and usage advice is intended to be a starting point to guide locally relevant policy development/action plans and transformative projects/programmes.

Specific combinations of indicators and contextual site information are also considered useful. One example is LUC + LPD + slopes greater than 15%. This map may be useful in identifying more sensitive areas in terms of land degradation in the selected watershed. It represents the areas where the land productivity dynamics are declining or stable but stressed (LPD values from 1 to 3) together with the places where the slope is more than 15%. This approach allows a user to select the plots/parcels where a focus on land restoration would have the highest immediate returns. This selection can be made at the LUC sectoral level of forest, agriculture (croplands) and livestock (grassland). However, LPD default data has limits in its use for this approach at plot/parcel level because of the low resolution of 1 km. If countries have already national LPD data at a finer spatial resolution of less than 1 km, it is recommended to examine these layers before finalising the planning of transformative projects.

Furthermore, a specific breakdown of the area of degradation within each LUC category per selected watershed is provided (e.g. Table 3). This table may also assist in identification of watershed specific or cross watershed land use policy.

It is important to note that the three suggested indicators (LUC, LPD and SOC) cannot possibly capture all land that is in a degraded state. They also cannot capture perfectly the degradation drivers that they do represent, as differences in resolution, timing of data acquisition and variance within methods, will always introduce some noise.

Taken together as well as separately and in combination with other contextual information, these indicators can highlight LDN hot spots where land degradation has been persistently active during the reference or baseline period, more or less from the late 1990s to 2010. Subsequently, degradation that has occurred prior to this period but has not been further degraded nor recovered, is not indicated in this assessment.

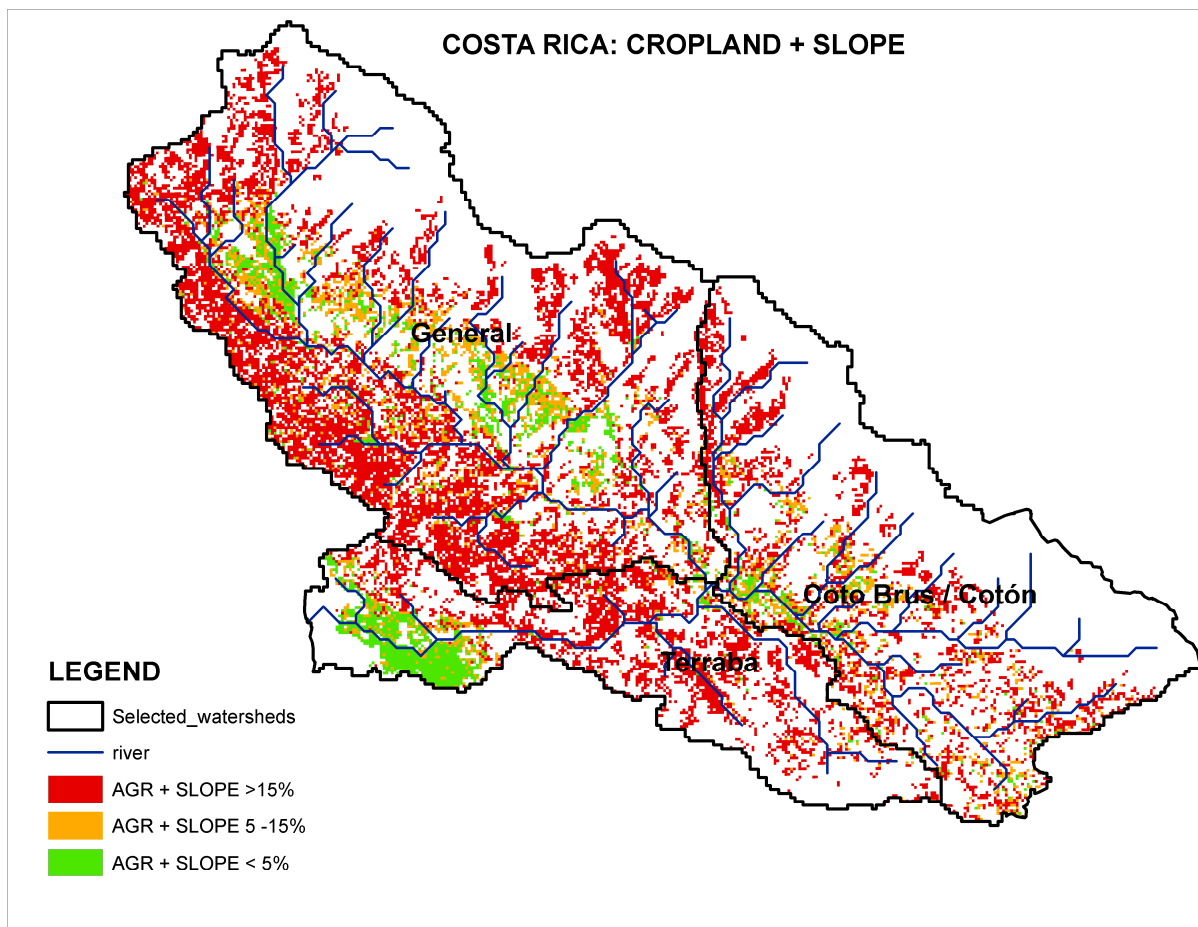


Figure 4. Cropland land cover combined with the different slope % ranges considered in the analysis. From green to red colour the degradation risk is increasing.

Priority	Watersheds	Degrading area: LPD + SOC (sq km)*	Cumulative sum degraded area (%)	Watershed area (sq km)	Cumulative sum national area (%)	SOC loss (tons)	Area SOC loss (sq km)**	Cumulative sum national SOC loss (%)	LPD 1-3 area (sq km)***	Cumulative sum LPD 1-3 (%)	Priority
	General	611	14%	2531	5%	-2376	0.8	1%	610	14%	
	Artificial areas remaining Artificial areas	4	0%	9	0%	0	0.0	0%	4	0%	
	Cropland remaining Cropland	352	8%	1195	2%	0	0.0	0%	352	8%	
	Forest remaining Forest	190	4%	1129	2%	0	0.0	0%	190	4%	
	Forest to Cropland	1	0%	1	0%	-2376	0.8	1%	0	0%	
	Forest to Shrubs, grasslands and sparsely vegetated areas	0	0%	1	0%	0	0.0	0%	0	0%	
	Shrubs, grasslands and sparsely vegetated areas remaining Shrubs, Wetlands remaining Wetlands	63	1%	195	0%	0	0.0	0%	63	1%	
		0	0%	1	0%	0	0.0	0%	0	0%	
	Coto Brus / Coto	338	22%	1619	8%	-6201	1.7	5%	337	22%	
	Cropland remaining Cropland	57	9%	418	3%	0	0.0	0%	57	10%	
	Forest remaining Forest	259	10%	1028	4%	0	0.0	0%	259	10%	
	Forest to Cropland	2	0%	2	0%	-6201	1.7	5%	0	0%	
	Forest to Shrubs, grasslands and sparsely vegetated areas	0	0%	1	0%	0	0.0	0%	0	0%	
	Shrubs, grasslands and sparsely vegetated areas remaining Shrubs, Wetlands remaining Wetlands	20	2%	170	1%	0	0.0	0%	20	2%	
		0	0%	0	0%	0	0.0	0%	0	0%	
	Turkey / Veracruz / Colorado	312	29%	2144	12%	-28080	7.4	19%	305	29%	
	Artificial areas remaining Artificial areas	127	3%	287	1%	0	0.0	0%	127	3%	
	Bare land and other areas remaining Bare land and other areas	0	0%	0	0%	0	0.0	0%	0	0%	
	Cropland remaining Cropland	58	11%	586	4%	0	0.0	0%	58	11%	
	Forest remaining Forest	76	12%	971	6%	0	0.0	0%	76	12%	
	Forest to Cropland	8	0%	7	0%	-28080	7.4	19%	1	0%	
	Forest to Shrubs, grasslands and sparsely vegetated areas	2	0%	14	0%	0	0.0	0%	2	0%	
	Shrubs, grasslands and sparsely vegetated areas remaining Shrubs, Wetlands remaining Wetlands	41	3%	265	1%	0	0.0	0%	41	3%	
		0	0%	13	0%	0	0.0	0%	0	0%	
	Terraba	294	36%	884	14%	-1260	0.5	20%	294	36%	
	Artificial areas remaining Artificial areas	0	3%	3	1%	0	0.0	0%	0	3%	
	Bare land and other areas remaining Bare land and other areas	1	0%	1	0%	0	0.0	0%	1	0%	
	Cropland remaining Cropland	133	14%	379	5%	0	0.0	0%	133	14%	
	Forest remaining Forest	122	15%	353	7%	0	0.0	0%	122	15%	
	Forest to Cropland	0	0%	0	0%	-1260	0.5	20%	0	0%	
	Forest to Shrubs, grasslands and sparsely vegetated areas	0	0%	0	0%	0	0.0	0%	0	0%	
	Shrubs, grasslands and sparsely vegetated areas remaining Shrubs, Wetlands remaining Wetlands	27	3%	123	1%	0	0.0	0%	27	4%	
		11	0%	24	0%	0	0.0	0%	11	0%	
	Tempisque	243	41%	3374	21%	0	0.0	20%	243	42%	
	Artificial areas remaining Artificial areas	5	3%	17	1%	0	0.0	0%	5	3%	
	Bare land and other areas remaining Bare land and other areas	0	0%	1	0%	0	0.0	0%	0	0%	
	Cropland remaining Cropland	138	17%	2113	9%	0	0.0	0%	138	17%	
	Forest remaining Forest	46	16%	393	8%	0	0.0	0%	46	16%	
	Shrubs, grasslands and sparsely vegetated areas remaining Shrubs, Wetlands remaining Wetlands	54	5%	843	3%	0	0.0	0%	54	5%	
		0	0%	8	0%	0	0.0	0%	0	0%	
	Grand Total	1798		10552		-37917	10.4		1788		

Table 3: specific breakdown of deeper analysis date per LUC category for default priority watersheds